

PROJECT TITLE:

TELCO CUSTOMER CHURN ANALYSIS

DATASET:

IBM'S TELCO CUSTOMER CHURN

BY:

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PROJECT OVERVIEW:

This project explores a telecommunications customer churn dataset using SQL to uncover patterns in customer behavior, service adoption, customer value, revenue generation, and churn.

My objective is to generate data-driven insights that can help the business improve customer retention, strengthen revenue performance, and support strategic decision-making.

BUSINESS PROBLEM:

Over time, customer churn has remained one of the most significant challenges facing telecommunications companies. Businesses are losing valuable customers, reducing recurring revenue, and increasing customer acquisition costs, all due to poor evaluation of customers' satisfaction.

Acquiring new customers is even more expensive than retaining existing ones, making customer retention essential for sustaining long-term profitability.

It is therefore, important for businesses to tackle hurdles faced by their customers, so as to reduce customer attrition and satisfy consumers, but how is that even possible when the businesses don't have a clear understanding of the factors that drive customer churn?

The telecommunications company represented in this dataset lacks clear visibility into the behaviors associated with customer churn. It is therefore necessary to identify the customer segments most likely to leave, determine how factors such as contract type, tenure, internet service, pricing, and value-added services influence churn, and assess the financial impact of customer attrition.

This analysis addresses these challenges by exploring customer behavior and revenue patterns using SQL-based exploratory data analysis. The insights generated provided evidence-based recommendations to support customer retention strategies, improve service offerings, and protect long-term revenue.

PROJECT SUMMARY:

This project analyzes a telecommunications customer churn dataset containing **7,032 customer records** to identify the key factors influencing customer attrition and their impact on business performance. SQL was used to clean, transform, and analyze the data through exploratory data analysis (EDA), focusing on customer characteristics, service adoption, churn patterns, and revenue performance.

The analysis revealed an overall churn rate of **26.58%**, indicating that approximately one in four customers discontinued their services. Customers on **month-to-month contracts** exhibited the highest likelihood of churn (42.71%), while those on one-year and two-year contracts showed substantially lower churn rates. Customers within their **first 12 months** also experienced the highest churn, suggesting that customer retention challenges are most significant during the early stages of the customer lifecycle.

Service type was another key factor influencing churn. Although **fiber optic** customers generated the highest total revenue, they also recorded the highest churn rate (41.89%), making this segment both the company's most valuable and most vulnerable. Conversely, customers subscribed to value-added services such as **Online Security, Online Backup, Device Protection, Tech Support, and Streaming TV** consistently demonstrated stronger retention than those who did not use these services.

Revenue analysis further showed that retained customers contributed **82.17%** of the company's historical revenue, while churned customers accounted for **17.83%**, representing approximately **\$2.86 million** in lost revenue. These findings highlight the financial impact of customer attrition and emphasize the importance of proactive retention strategies.

Based on these insights, the report recommends prioritizing retention initiatives for month-to-month and fiber optic customers, encouraging migration to longer-term contracts, strengthening customer engagement during the first year of service, and promoting value-added services that are associated with lower churn. Implementing these strategies could improve customer retention while protecting the company's most valuable revenue streams.

DATASET OVERVIEW

The dataset used in this project is the IBM Telco Customer Churn dataset, which contains information on 7,032 customer records after data cleaning. It includes customer demographics, subscription details, billing information, service usage, contract characteristics, and churn status.

The primary objective of the dataset is to support the analysis of customer behavior and identify factors associated with customer churn. Each record represents an individual customer and includes attributes that describe their relationship with the company.

The dataset consists of the following categories of information:

- **Customer Information:** Customer ID, tenure, and Senior Citizen status.
- **Services Subscribed:** Internet Service, Online Security, Online Backup, Device Protection, Tech Support, and Streaming TV.
- **Account Information:** Contract type, Payment Method, Monthly Charges, and Total Charges.
- **Target Variable:** Churn, indicating whether a customer remained with the company or discontinued its services.

Before analysis, the dataset was cleaned to improve data quality. The cleaning process involved handling missing values, correcting inconsistent data entries, standardizing categorical values, and removing duplicate records. These preprocessing steps ensured the dataset was suitable for reliable exploratory data analysis.

REPORT

1. Total Number of Customers

Business Question

How many customer records are available for analysis?

Findings

The dataset contains **7,032 customer records** after cleaning, providing a comprehensive sample for analyzing customer demographics, service subscriptions, revenue patterns, and churn behavior. This size is sufficient to identify meaningful trends and support data-driven business decisions.

2. Customer Adoption Across Contract Types

Business Question

Which contract type is most widely adopted by customers?

Findings

Month-to-month contracts account for the largest share of the customer base, indicating that customers generally prefer the flexibility of short-term commitments to long-term contracts.

This distribution suggests that a significant portion of the customer base may be more susceptible to switching providers due to the absence of long-term obligations.

Recommendation

- Introduce targeted incentives, such as discounted rates or loyalty rewards, to encourage customers on month-to-month contracts to migrate to one-year or two-year plans.

3. Customer Adoption Across Internet Services

Business Question

Which internet service is most widely adopted by customers?

Findings

Fiber optic is the most widely subscribed internet service, with 3,096 customers, representing approximately 44.0% of the customer base.

DSL follows with 2,416 customers (34.3%), while 1,520 customers (21.6%) do not subscribe to any internet service.

The strong adoption of Fiber optic indicates that it is the company's flagship internet offering and the primary driver of its internet customer base. However, its large customer base also means that changes in customer satisfaction or service quality within this segment are likely to have a disproportionate impact on overall business performance.

Recommendations

- Given that Fiber optic serves nearly half of all customers, maintaining service quality within this segment should be a strategic priority.
- Regular monitoring of customer satisfaction, network performance, and service reliability is essential to protect the company's largest customer segment and minimize future customer losses.

4. Customer Adoption Across Payment Methods

Business Question

Which payment methods are most commonly used by customers?

Findings

Electronic check is the most frequently used payment method, with 2,365 customers, representing approximately 33.6% of the customer base.

This is followed by Mailed check (1,604 customers, 22.8%), Bank transfer, automatic (1,542 customers, 21.9%), and Credit card, automatic (1,521 customers, 21.6%).

The relatively balanced adoption of the three remaining payment methods suggests that customer preference is concentrated around electronic check, while the other options are utilized at comparable levels. Given that electronic check is the dominant payment method, any issues associated with this group have the potential to affect a substantial portion of the customer base.

5. Customers by Tenure Group

Business Question

How is the customer base distributed across different stages of the customer lifecycle?

Findings

The customer base is relatively well distributed across all tenure groups, with 31.8% of customers having remained with the company for 49 months or more, making this the largest customer segment.

Customers with 0–12 months of tenure closely follow with 30.9%, indicating a strong inflow of recently acquired customers.

The remaining customer base consists of 1,594 customers (22.6%) with 25–48 months of tenure and 1,024 customers (14.5%) with 13–24 months of tenure.

The presence of a large long-term customer segment indicates strong customer retention power.

Recommendation

- Develop retention strategies that address customers at different stages of their lifecycle.
- Newly acquired customers may benefit from stronger onboarding and engagement initiatives, while long-term customers can be retained through loyalty programs and personalized service offerings.

6. Average Monthly Charges (Average Monthly Revenue per Customer)

Business Question

What is the average monthly revenue generated from each customer?

Findings

The company generates an average of \$64.80 in monthly revenue per customer.

7. Average Monthly Charges by Contract

Business Question

How does average monthly revenue vary across different contract types?

Findings

Customers on Month-to-month contracts generate the highest average monthly revenue at \$66.40, followed closely by One-year contracts (\$65.08). Customers on Two-year contracts have the lowest average monthly charge at \$60.87.

This finding signifies that customers on two-year contracts pay slightly less on average, likely reflecting discounted long-term pricing designed to encourage customer commitment.

Despite generating the lowest monthly revenue, two-year customers also experience the lowest churn, indicating that long-term retention may compensate for lower monthly charges through greater customer lifetime value.

Recommendation

- Continue using competitive pricing to encourage customers to adopt longer-term contracts

8. Average Monthly Charges by Internet Service

Business Question

How does average monthly revenue differ across internet service categories?

Findings

Fiber optic customers generate the highest average monthly revenue at \$91.50, substantially exceeding DSL customers (\$58.09) and customers without internet service (\$21.08). Fiber optic customers therefore contribute significantly more monthly revenue than any other internet service segment.

Recommendation

- Retaining Fiber optic customers should be a strategic priority.
- Improving service reliability, customer support, and perceived value within this segment can help protect the company's highest source of recurring monthly revenue while reducing the financial impact of churn.

9. Average Monthly Charges by Churn

Business Question

Is there a difference in monthly spending between customers who churned and those who remained with the company?

Findings

Customers who churned paid an average monthly charge of \$74.44, compared with \$61.31 for customers who remained with the company. On average, churned customers spent approximately 21% more per month than retained customers.

This finding suggests that customers generating higher recurring revenue are also more likely to leave the company.

Recommendation

- Since customers paying premium monthly fees are likely to have higher expectations regarding service quality, reliability, and customer support, they should be closely monitored and ensured to enjoy targeted perks and retention initiatives before they churn.

10. Churn Rate by Monthly Charge Segment

Business Question

Which monthly pricing segment experiences the highest customer churn rate?

Findings

Customers paying **\$70 or more per month** exhibit the highest churn rate at **35.50%**, considerably higher than customers in the **Medium (\$35–\$69.99)** segment (**23.69%**) and more than three times the churn rate of customers in the **Low (\$0–\$34.99)** segment (**10.90%**).

Recommendation

- Customers within the highest pricing segment should be considered a priority for retention initiatives. The company should evaluate whether premium plans consistently deliver sufficient value through service quality, network performance, and customer support.
- Regular customer satisfaction surveys and targeted retention offers may help reduce churn within this high-value segment.

11. Average Total Charges by Contract

Business Question

How does customer lifetime revenue vary across different contract types?

Findings

Customers on **two-year contracts** generate the highest average lifetime revenue at **\$3,728.93**, followed by **one-year contracts (\$3,034.68)**. In comparison, customers on **month-to-month contracts** generate an average of only **\$1,369.25** over their relationship with the company.

Recommendation

- Encouraging customers to transition from month-to-month plans to longer-term contracts should remain a strategic priority.

12. Top 10 Customers by Total Revenue

Business Question

Which customers have generated the highest total revenue for the business?

Findings

The top ten customers have each generated between **\$8,477.70** and **\$8,684.80** in total revenue while maintaining consistently high monthly charges ranging from **\$114.90** to **\$118.75**.

These customers represent some of the company's highest-value relationships. Losing even a small number of customers from this group would result in a significantly large revenue impact compared to the average customer.

Recommendation

- Develop personalized loyalty rewards to protect these customers.

13. Overall Churn Rate

Business Question

What proportion of customers have discontinued their services with the company?

Findings

Approximately **27% of customers have churned**, indicating that more than one in four customers have left the company.

A churn rate of this magnitude can significantly affect recurring revenue, customer lifetime value, and long-term business growth.

Recommendation

- Retention should be an utmost business priority at this point.

14. Average Tenure by Churn

Business Question

How long do churned customers remain with the company compared with retained customers?

Findings

On average, retained customers stay with the company for **approximately 20 months longer** than churned customers.

This means that the risk of churn is highest during the earlier stages of the customer relationship and declines as customers stay with the company for longer periods.

Recommendation

- Strengthen customer onboarding and engagement during the first year of service.

15. Churn Rate by Contract Type

Business Question

Which contract type has the highest likelihood of customer churn?

Findings

Customers on **Month-to-month contracts** have a churn rate of **42.71%**, meaning that **nearly one out of every two** customers on this contract type eventually leaves the company. This is significantly higher than the churn rate for **one-year contracts (11.28%)** and **two-year contracts (2.85%)**.

In other words, month-to-month customers are **almost four times more likely to churn than one-year customers** and **approximately fifteen times more likely to churn than two-year customers**.

Recommendation

- Prioritize retention efforts for month-to-month customers by offering incentives to migrate to longer-term contracts.

16. Churn Rate by Internet Service

Business Question

Which internet service category has the highest customer churn rate?

Findings

Customers using Fiber optic internet service exhibit the highest churn rate at 41.89%, more than double the churn rate of DSL customers (19.00%) and significantly higher than customers without internet service (7.43%).

This simply signifies that Fiber optic customers are more likely to leave the company despite subscribing to a premium internet service.

Recommendation

- Conduct a detailed investigation into the Fiber optic customer segment to identify the primary drivers of churn.
- Review service reliability, network performance, and customer support quality for Fiber optic users.

17. Churn Rate by Tenure Group

Business Question

At which stage of the customer lifecycle is churn most likely to occur?

Findings

Customers with 0–12 months of tenure exhibit the highest churn rate with a 47.68% churn rate, indicating that nearly half of all newly acquired customers leave within their first year, while churn decreases steadily as tenure increases.

Long-term customers are significantly less likely to leave, meaning that retention improves as the customer relationship matures.

Recommendation

- Focus retention initiatives on newly acquired customers during their first year with the company.
- Improve onboarding experiences, proactive customer support, and early engagement programs that can help reduce churn during this delicate stage.

18. Churn by Internet Service

Business Question

How are churned customers distributed across internet service categories?

Findings

Among customers who churned, Fiber optic accounts for the highest number of customer losses (1,297), compared with 459 DSL customers and only 113 customers without internet service.

Fiber optic therefore contributes the largest share of overall churn, reinforcing the earlier finding that this service experiences both the highest churn rate and the highest number of churned customers.

Recommendation

- Investigate the underlying causes of dissatisfaction and implement targeted retention initiatives within this group

19. Churn by Payment Method

Business Question

How is customer churn distributed across different payment methods?

Findings

Customers using **Electronic check** account for the highest number of churned customers (**1,071**), substantially exceeding customers paying by **Mailed check (308)**, **Bank transfer (automatic) (258)**, or **Credit card (automatic) (232)**.

The huge figure might also be a result of the fact that the electronic check is the most preferred payment method by customers.

Recommendation

- Perform further internal investigations to determine why Electronic check customers experience higher churn.

20. Churn by Value Added Services.

Business Question

How does the adoption of value-added services influence customer churn?

Findings

I've experienced a consistent pattern across Online Security, Online Backup, Device Protection, Tech Support, and Streaming TV.

Customers who subscribe to these value-added services generally experience lower levels of churn than customers who do not subscribe.

Among these services, Online Security and Online Backup demonstrate the clearest relationship with customer retention. Customers without these services account for substantially more churned customers than those who subscribe to them.

Similar patterns are observed for Device Protection, Tech Support, and Streaming TV, suggesting that customers who engage with multiple services tend to remain with the company longer.

This is simply as a result of the fact that, as customers rely on more services, the perceived cost and inconvenience of switching providers may increase, making them less likely to churn.

Recommendation

- Encourage greater adoption of value-added services through bundled packages and personalized recommendations based on customer needs.

21. Total Revenue by Internet Service

Business Question

Which internet service generates the highest total revenue for the company?

Findings

Fiber optic customers generated approximately **\$9.92 million** in total revenue, nearly double the **\$5.12 million** generated by DSL customers and substantially more than customers without internet service (**\$1.01 million**). This indicates that fiber optic is the company's most valuable internet offering from a revenue perspective.

Recommendation

Although fiber optic is the largest revenue contributor, it also records the highest churn rate (41.89%). The business should prioritize retaining fiber optic customers through service quality improvements and better customer support.

22. Total Revenue by Churn Status

Business Question

How is total revenue distributed between retained customers and customers who have churned?

Findings

Customers who remained with the company generated approximately **\$13.19 million** in total revenue, while customers who eventually churned contributed **\$2.86 million**.

Although retained customers account for the majority of revenue, the revenue generated by customers who later left the company is still substantial, demonstrating that churn affects long-term revenue performance.

Recommendation

- Highly prioritize customer retention.

23. Revenue Share by Churn Status (%)

Business Question

What percentage of the company's total historical revenue came from customers who churned compared to those who were retained?

Findings

Retained customers account for **82.17%** of the company's total historical revenue, while customers who eventually churned contributed **17.83%**.

Although only about **18%** of historical revenue has been lost, this still leaves a huge financial impact on revenue and tells us the importance of satisfying customers before they leave.

Recommendation

- Focus retention efforts on customer segments with both high churn rates and high revenue potential, particularly fiber optic and month-to-month customers, because a modest reduction in churn within these segments could preserve a great share of future revenue.

KEY FINDINGS

1. Customer churn remains a significant business challenge.
2. Month-to-month customers are the most vulnerable to churn.
3. Customer churn is concentrated during the early stages of the customer lifecycle.
4. Fiber optic customers represent the company's most valuable users yet highest-risk customer segment, generating the highest total revenue while also recording the highest churn rate (41.89%).
5. Higher monthly charges are associated with increased churn.
6. Value-added services appear to support customer retention.
7. Electronic check customers contribute the largest volume of customer churn.
8. Customer retention directly influences long-term revenue.

FINAL BUSINESS RECOMMENDATIONS

1. Reduce churn among month-to-month customers by introducing incentives to migrate to one- and two-year contracts.
2. Prioritize retention of fiber optic customers through improvements in service quality and proactive customer support.
3. Promote value-added services (Online Security, Online Backup, Device Protection, Tech Support, and Streaming TV), as customers subscribing to these services consistently demonstrate lower churn.
4. Identify customers within their first 12 months and implement targeted onboarding and engagement campaigns, since this group experiences the highest churn.
5. Focus retention efforts on high-value customers, as preserving existing revenue is more cost-effective than replacing lost customers.

CONCLUSION

Overall, this analysis shows that customer churn is driven less by customer demographics and more by contract commitment, service type, tenure, and engagement with value-added services.

Addressing these factors would improve customer retention and also protect the company's highest-value revenue streams, supporting stronger long-term profitability.